Landau's Theoretical Minimum had included problems from *all* the branches of mathematics that Landau deemed important for theoretical physics. Topology was not among them. Calculus, complex variables, the qualitative theory of differential equations, group theory, and differential geometry were all covered; they would all be needed in a physicist's career. But topology would not be needed. Landau had nothing against topology; he just ignored it; it was irrelevant—and his view of its irrelevance became near gospel among most Soviet theoretical physicists in the 1940s through the 1960s.

This view was transmitted to theoretical physicists around the world by the set of textbooks, called *Course of Theoretical Physics*, that Landau and Lifshitz wrote. These became, worldwide, the most influential set of physics texts of the twentieth century, and like Landau's Theoretical Minimum examinations, they ignored topology.

Curiously, topological techniques were introduced into relativity research in an abortive way, long before Penrose's theorem, by two Soviet mathematicians in Leningrad: Aleksander Danilovich Aleksandrov and Revol't Ivanovich Pimenov. In 1950–59, Aleksandrov used topology to probe the "causal structure" of spacetime, that is, to study the relationships between regions of spacetime that can communicate with each other and those that cannot. This was just the type of topological analysis that would ultimately pay rich dividends in the theory of black holes. Aleksandrov built up a rather powerful and beautiful topological formalism, and in the mid-1950s that formalism was picked up and pushed further by Pimenov, a young colleague of Aleksandrov's.

But in the end this research led nowhere. Aleksandrov and Pimenov had little contact with physicists who specialize in gravitation. Such physicists would have known what kinds of calculations were useful and what were not. They might have told Aleksandrov and Pimenov that the big bang singularity or gravitational implosion of stars deserved probing with their formalism. But no such advice was to be had in Leningrad; the key physicists worked 600 kilometers southeast of Leningrad, in Moscow, and were ignorant of topology and topologists. The Aleksandrov–Pimenov formalism flowered, and then went dormant.

Its dormancy was forced by the fates of Aleksandrov and Pimenov: Aleksandrov became the rector (president) of Leningrad University, and had inadequate time for further research. Pimenov was arrested in 1957 for